Ivan Katorgin

Born May 2, 1984

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Location: Izhevsk. Ready to relocate. Ready to business-trip occasionally.

Desired position

Expert in Oil and Gas Field Development, Domain Data Scientist (Oil and Gas)

Work experience

January 2024 – currently

Netology Educational Projects (part-time job)

Moscow

Data Scientist, Data Science Expert

Main responsibilities:

- Students Mentor of Analytics and Data Science.

Basic work results:

- Implemented 15+ end-to-end DS-projects in ML, DL, NLP, CV domains;
- Won 1st place at the Netology data analytics workshop;
- Finalist at the "Emkaton of Moscow Economics" hackathon;
- Participation in the Siam ML Hack hackathon on the application of ML methods in processing and interpreting well tests.

Examples of completed DS-projects:
1. AI Diagnostics of Brain Pathologies from CT Scans with Decision Visualization:
Stack: PyTorch, OpenCV, Grad-CAM;

- Developed and trained models (ResNet18, EfficientNet) for pathology classification from CT scans with up to 85% accuracy;
- Implemented attention mechanisms and Grad-CAM visualization for model decision interpretation;
- Executed full project lifecycle: from data quality analysis and augmentation to validation and preparation for clinical application.

2. Multi-Task Facial Attribute Classification (gender, age, ethnicity):
Stack TensorFlow, Keras, Transfer Learning, Pandas, NumPy, Scikit-learn;

- Designed a computer vision system for simultaneously solving 3 tasks (classification, regression) on a shared encoder;
- Implemented and compared 5 neural network architectures. The best model achieved 92.3% accuracy for gender and 85.7% for ethnicity, MAE of age 6.2 years;
- Applied attention mechanisms and multi-task learning to improve model generalization.

3. Attribute-Guided Face Generation using GANs:
Stack: TensorFlow, Keras;

- Developed 5 GAN architectures (Conditional GAN, DCGAN, DCGAN with Attention, hybrid models) for generating faces based on specified attributes (gender, age, ethnicity);
- Created a functional face generation system with attribute control;
- Analyzed clustering of generated images in the feature space.

4. Comparative Analysis of Attention Mechanisms in Seq2Seq Architecture for Machine Translation:
Stack: PyTorch, NLTK, Pandas, NumPy, Matplotlib, Seaborn;

- Implemented and compared 3 attention mechanisms (Scalar Dot-Product, Multi-Head, MLP-based) in an Encoder-Decoder architecture;
- Achieved BLEU-4 Score of 0.85 on English-Russian translation task;
- Visualized attention matrices to analyze model performance and conducted ablation studies.

5. Comprehensive Time Series Modeling: From ARIMA to Machine Learning:
Stack: Statsmodels, Arch, Scikit-learn, Seaborn;

- Developed a universal methodology for time series analysis and forecasting;
- Built and compared multiple models (ARIMA, SARIMAX, ARCH) for stationary and non-stationary series, implemented automatic parameter selection;
- Achieved MAE of 88 when forecasting test data using both statistical methods and linear regression on lags.

6. Credit Default Prediction for a Microfinance Company:
Stack: Pandas, NumPy, Scikit-learn, XGBoost, Seaborn;

- Developed a scoring system for predicting defaults with 99.13% accuracy and 0.99 AUC;
- Executed full cycle of feature engineering, data balancing, and comparative analysis of 4 ML models.

More details on these and other completed DS-projects can be found on my GitHub:
https://github.com/IvanKatorgin
and in my portfolio:
https://drive.google.com/file/d/1hXzgeacYzjR9r62yszqLjyuL5SCbtIJo/view?usp=sharing

April 2018 – currently

Izhevsk Petroleum Research Center (Rosneft Oil Company)

Izhevsk

Chief Specialist of Design Oil & Gas Fields Development Division

Main responsibilities:

- Monitoring compliance with license obligations for hydrocarbons and water;
- Geological and field development analysis (well testing analysis, production logging analysis, base production analysis, reservoir pressure maintenance analysis, reservoir stimulation, flow rate analysis, production and injection system analysis, reserve depletion analysis);
- Preparing proposals and measures to optimize the development system;
- Preparing reservoir stimulation programs;
- Preparing drilling rankings for new wells and sidetracks;
- Performing calculations and expert reviews of forecasted technological development indicators using various methods (material balance, analytics, hydrodynamic modeling);
- Preparing exploration programs, additional exploration, and research work;
- Calculating forecasted water-related indicators (placement, water intake);
- Field development monitoring;
- Reserves calculation (initial and remaining reserves);
- Reserves audit (PRMS, SEC), reserves inventory;

- Organizing and preparing project documentation for hydrocarbons, hydrogeology, surveying work;
- Preparing draft contracts, terms of reference, and commercial proposals;
- Ensuring compliance with regulatory requirements, completeness and quality of project documentation, and adherence to approval procedures;
- Ensuring timely submission of reports on geology and development issues;
- Ensuring timely execution of work according to the thematic plan and timely closure of work stages related to geology and development;
- Ensuring quality control and expert review of work in geology and development;
- Forming project teams, allocating resources for planned and ad-hoc tasks, coordinating project team activities;
- Approving and defending work before the Client and government authorities (State Reserves Committee, Central Development Commission);
- Interaction with government agencies (Ministry of Energy, Rosnedra) and contractors on geology and development issues.

Basic work results:

- Organized work on preparing project-technological documents for field development for RN-Yuganskneftegaz and RN-Purneftegaz;
- Prepared and defended project documents for RN-Yuganskneftegaz:

- Supplement to the technological development project of the Petelinskoye oil field, KhMAO-Yugra, Tyumen Region (annual oil production >1 million tons);

- Supplement to the technological development scheme of the Kinyaminskoye oil field, KhMAO-Yugra, Tyumen Region;

- Supplement to the technological development project of the Srednebalykskoye (main part) oil field, KhMAO-Yugra, Tyumen Region;

- Prepared over 20 project documents for the development of groundwater fields in various regions of Rosneft Oil Company's operations (RN-Yuganskneftegaz, Samaraneftegaz);
- Monitored water license compliance for Bashneft and participated in regular production meetings on hydrogeology with Bashneft;
- Completed surveying projects for ROSPAN INTERNATIONAL and other subsidiaries of Rosneft Oil Company.

August 2016 – December 2017

West Siberian Research Institute of Geology and Geophysics

Tyumen

Head of Complex Oil & Gas Fields Development Analysis Division

Main responsibilities:

- Planning, organizing, technical supervision, preparation and defense of reserve estimation reports and project-technological development documents before the Subsoil User's Scientific and Technical Council and various-level government commissions;
- Interaction at all stages of work with the Client and government authorities (State Reserves Committee, Central Development Commission);
- Forming working groups and allocating resources across projects and tasks;
- Supporting contractual activities for geology and development work;
- Participating in procurement for geological and geophysical study of subsoil assets, licensing, and hydrocarbon field development;
- Ensuring timely execution of work and timely closure of work stages related to geology and development;
- Ensuring quality control and expert review of work in geology and development;

- Ensuring compliance with regulatory requirements, completeness and quality of documentation, and adherence to approval procedures;
- Subsoil use monitoring and analysis of the current resource base of fields;
- Conducting expert reviews of forecasted technological development indicators and selecting optimal field development strategies;
- Formulating reservoir stimulation and well workover programs based on analysis, modeling, geological and field data, etc.

Basic work results:

- Supplement to the technological development scheme of the Radishchevsko-Gartovskoye oil and gas field, Saratov Region;
- Trial operation project for exploration well No. 378 of the Urengoyskoye oil and gas condensate field;
- Trial operation project for exploration well No. 507 of the Beregovoye oil and gas condensate field.

February 2015 – June 2016

Tyumen Petroleum Research Center (Rosneft Oil Company)

Tyumen

Deputy Director of Department, Department Manager, Department of Geology and Development of Varyegannefteqaz Fields

Main responsibilities:

- Monitoring compliance with license obligations;
- Conducting expert reviews of forecasted technological development indicators;
- Assessment and analysis of risks (geological and economic) and uncertainties in field development and new well drilling;
- Evaluating development indicators and economic efficiency of new oil and gas assets;
- Participating in the preparation of integrated development projects for the supervised region;
- Participating in the formation of rankings and schedules for new well drilling;
- Ensuring timely execution of work according to the thematic plan and timely closure of work stages related to geology and development;
- Ensuring quality control and expert review of work in geology and development;
- Planning, organizing, preparing and defending reserve estimation reports and project-technological development documents before the Subsoil User's Scientific and Technical Council and various-level government commissions;
- Interaction with government authorities (State Reserves Committee, Central Development Commission) and contractors on geology and development issues;
- Preparing draft contracts, terms of reference, and work schedules for research and design work in geology and development, calculating work costs;
- Supporting contractual activities for geology and development work;
- Ensuring compliance with regulatory requirements, completeness and quality of documentation, and adherence to approval procedures;
- Ensuring timely submission of reports on geology and development issues.

Basic work results:

- Organized work with contractors to prepare project-technological documents for the Varyegannefteqaz Reserve Management Department;
- Prepared an integrated development project for the operating region of RN-Nyaganneftegaz (annual oil production >6 million tons, active well stock >4,000);

- Prepared and defended project documents for Varyeganneftegaz:

- Supplement to the technological development scheme of the Bakhilovskoye oil and gas condensate field, KhMAO-Yugra, Tyumen Region;

- Supplement to the development project of the Permyakovskoye oil field, KhMAO-Yugra, Tyumen Region;

- Supplement to the technological development scheme of the Okunevskoye oil field, KhMAO-Yugra, Tyumen Region;

- Trial operation project for well No. 5P of reservoir YuV5 of the Severo-Varyeganskoye oil and gas condensate field.

April 2007 – February 2015

RN-SakhalinNIPImorneft (Rosneft Oil Company)

Yuzhno-Sakhalinsk

Chief Project Engineer, Assistant Chief Project Engineer, Leading Specialist, Category 1 Engineer, Department of Geology and Development

Main responsibilities:

- Planning, organizing, preparing and defending reserve estimation reports and project-technological development documents before the Subsoil User's Scientific and Technical Council and various-level government commissions;
- Interaction with government authorities (State Reserves Committee, Central Development Commission) and contractors on geology and development issues;
- Preparing draft contracts, terms of reference, and work schedules for research and design work in geology and development, calculating work costs;
- Supporting contractual activities for geology and development work;
- Ensuring compliance with regulatory requirements, completeness and quality of documentation, and adherence to approval procedures;
- Ensuring timely submission of reports on geology and development issues;
- Recruiting new employees, personnel development, and mentoring;
- Preparing, implementing, and updating local regulatory documents within area of competence.

Basic work results:
Throughout the entire work period, prepared and defended over 10 project-technological field development documents and reserve estimation reports for onshore and offshore Sakhalin, including:

- Development project for the Odoptu-sea field (North Cupola);

- Supplement to the technological development scheme of the Gilyako-Abunan field;

- Supplement to the development project of the Mukhto field;

- Supplement to the technological development scheme of the Severnaya Okha field;

- Supplement to the development project of the Odoptu field;

- Development project for the Ekhabi field;

- Reserves recalculation and feasibility study for oil recovery factor of the Ekhabi field;

- Technological development scheme of the Kydylanyi field;

- Development project for the Tungor field, and other projects and reports.

Awards:
- Received honorary certificate from RN-SakhalinNIPImorneft;

- Diploma "Young Specialist of the Year" of Rosneft Oil Company.

October 2005 – April 2007

Service Nafta

Moscow

Reservoir Engineer of Reserve Estimation Division

Main responsibilities:

- Analysis of oil & gas fields development;
- Geological modeling and Reservoir simulation;
- Writing sections of reserves estimation reports.

Education

2025

Netology

Data Scientist: Advanced track

2013

Gubkin Russian State University of Oil & Gas

Economics and Management at Oil & Gas Enterprises

2006

Gubkin Russian State University of Oil & Gas

Petroleum Geology

Professional development, courses

2026

MLOps

Netology

2023

Data Visualization: from boring charts to interactive dashboards

Netology

2022

Work-life balance and strategic thinking

H Alfa Digital

2020

Personal effectiveness and management skills

City business School

2014

International internship in "Kaizen" in Tokyo, Japan

Ministry of Foreign Affairs of Japan

2013

Management in Business

International Management Institute LINK

2013

Presidential Training Program for Management Personnel for Organizations of the National Economy of the Russian Federation

International Management Institute LINK

2012

Modern Mentoring Tools in Production

MAINSTREAM

2012

Schlumberger Technological Seminars

Schlumberger

2010

New in Drilling and Modern Methods of Formation Penetration, Completion, and Near-Wellbore Zone Treatment

Oil and Gas Business Institute

2009

Business Presentations, Public Speaking

Russian-American Business Training Center

2008

Internal Audits of Quality Management Systems according to ISO 9001:2000

Bureau Veritas

Skills

Languages

Russian – Native
English – C1 – Advanced

Skills

Petroleum geology and field development Field development analysis Well testing analysis Production logging analysis Drilling Sidetracks Reservoir stimulation Hydraulic fracturing Geological modeling Hydrodynamic modeling Project-technological documentation Reserves calculation Hydrogeology Exploration Eclipse Tempest tNavigator Rexlab Petrel Irap RMS NGT Smart RN-KIN Baspro Kappa Python Pandas Numpy SciPy PyTorch TensorFlow Keras Scikit-learn XGBoost LightGBM Statsmodels NLTK OpenCV SQL MySQL PostgreSQL MongoDB Matplotlib Seaborn Plotly Tableau Power BI Git PySpark Deep Learning Drafting technical specifications

Additional information

About me

Experienced manager and expert in geology and field development with 20+ years of experience. Retrained as a Data Scientist to apply modern ML/DL methods to solve complex oil and gas industry problems. Combine deep domain expertise (prepared and defended 40+ project-technological documents before the Central Development Commission/State Reserves Committee) with strong DS skills (15+ end-to-end DS-projects in ML, DL, NLP, CV). Aimed at implementing data-driven solutions and creating digital twins of fields. Awarded a Certificate of Honor from RN-SakhalinNIPImorneft and a Diploma from Rosneft Oil Company.

More details on completed oil & gas projects can be found here:
https://drive.google.com/file/d/1dBAjwQnYsRqQV0Wgv0raAVJ91xWrC2Hi/view?usp=sharing

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https://github.com/IvanKatorgin
and in my portfolio:
https://drive.google.com/file/d/1hXzgeacYzjR9r62yszqLjyuL5SCbtIJo/view?usp=sharing